



CONCEPTUAL BIOLOGY AND EARTH AND SPACE SCIENCE

ACADEMIC

Grade 12
Curriculum Guide

CONCEPTUAL BIOLOGY AND EARTH AND SPACE SCIENCE

Prerequisite: General Science (Core Subject)

Course Description:

This course develops the learners' understanding of fundamental scientific concepts by connecting them to real-life experiences, community contexts, and everyday decision-making. The learners explore heredity, biodiversity, and biotechnology to understand how genetic variation influences observable traits, how biodiversity supports ecosystems and community life in the Philippines, and how biotechnology applications improve health, food production, and overall well-being. Emphasis is placed on recognizing patterns, explaining biological phenomena, and designing simple, community-based solutions.

The course also examines natural resources, environmental sustainability, human impacts, climate change, and space technologies. The learners analyze how human activities affect the environment, understand the mechanisms of climate change, and explore how satellites and other space technologies provide data that help monitor Earth and improve daily life. Overall, the course highlights the role of science in addressing real-world problems and empowers learners to apply scientific knowledge in proposing practical solutions for their communities.

Academic: Elective

Time Allotment: 80 hours for one term

CONTENT	CONTENT STANDARDS <i>The learners demonstrate an understanding that...</i>	LEARNING COMPETENCIES <i>The learners...</i>
Conceptual Biology		
1. Heredity and Genetic Variation	1. traits are passed from parents to offspring through genes, and that variations in DNA result in observable differences among individuals and populations;	1. identify patterns of inherited traits in familiar human characteristics (e.g., dimples, widow's peak, attached or free earlobes, and other inherited traits) using simple family examples or pedigrees to explain that genes are passed from parents to offspring;
		2. explain that differences in physical characteristics among people (e.g., skin tone, height, hair type and other

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		inherited traits) are features of genetic diversity resulting from variations in DNA;
2. Biodiversity and Conservation in the Philippines	2. natural variation in inherited traits contributes to biodiversity, and that endemic organisms play important roles in maintaining ecosystems and supporting community life; and	3. explain that natural variation in inherited traits contributes to the variety of plants and animals found in the Philippines;
		4. describe endemic organisms in the Philippines by identifying examples found in their locality or nearby regions and explaining their importance to local ecosystems and communities;
		5. design a community-based activity that helps protect and preserve local biodiversity and describe how it can be applied in the community;
3. Biotechnology in Daily Life	3. biotechnology, supported by digital tools for analyzing biological information, is applied in everyday life to improve human health, food production, and community well-being.	6. describe the importance and everyday uses of traditional biotechnology (e.g., fermentation in food production, preservation, and household practices);
		7. describe how biotechnology helps improve crop production and nutritional value to support food needs;
		8. explain how biotechnology, supported by digital tools for analyzing biological information, help scientists develop medicines, vaccines, and medical technologies that address health problems in communities; and
		9. explain how the responsible and ethical use of biotechnology can support community well-being while minimizing risks to health, society, and the environment.

CONTENT	CONTENT STANDARDS <i>The learners demonstrate an understanding that...</i>	LEARNING COMPETENCIES <i>The learners...</i>
Performance Standard <i>The learners can design practical and simple solutions or awareness activities that demonstrate understanding of heredity, biodiversity, and biotechnology and their applications in improving everyday life in the community.</i>		
Suggested Performance Tasks • Create a family trait chart or simple pedigree showing inherited characteristics. • Develop a poster or infographic showing examples of endemic species in the local area and their importance. • Design a community-based biodiversity protection plan (e.g., school garden, waste management, tree planting proposal). • Create an informational campaign material (poster, brochure, or short video) explaining how vaccines, medical technologies, or improved crops help communities.		
Conceptual Earth and Space Science		
4. Natural Resources and Environmental Sustainability	4. natural resources are essential for daily living and that sustainable practices help reduce environmental impact and ensure resource availability for present and future needs;	10.explain how conserving natural resources (e.g., water, energy, raw materials) helps reduce environmental damage, maintain resource availability, and support daily needs such as food production, energy use, and clean water supply;
		11.identify ways to practice environmental sustainability in everyday life and describe how these actions (e.g., waste reduction, energy and water conservation, use of renewable energy, and sustainable food practices) help protect the environment;
5. Human Activities and Climate Change	5. human activities release pollutants and greenhouse gases that contribute to climate change, which affects ecosystems and	12.explain how human activities such as fossil fuel burning, deforestation, industrial processes, and mining contribute to climate change by releasing greenhouse gases and other pollutants that affect the environment;

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	communities at local and global levels;	13.describe how greenhouse gases trap heat in the atmosphere and how increased greenhouse gas levels enhance the greenhouse effect, leading to global warming; 14.use information from reliable data sources (e.g., PAGASA reports, satellite data, climate graphs, and online databases) to analyze how climate change affects local communities and ecosystems, and relate these effects to global phenomena such as sea-level rise and increased frequency of extreme weather events;
6. Earth Observation and Space Technologies	6. space technologies, including satellites and space probes, provide important data that help scientists understand Earth's systems, environmental changes, and the universe;	15.identify Filipino and international contributions to space technologies and describe how these advancements benefit society (e.g., communication, weather forecasting, disaster monitoring); 16.use information from satellite images, weather maps, and online databases (e.g., PAGASA, NASA, PhilSA) to explain how satellites collect and provide data about Earth's weather, climate, and environment; 17.explain how robotic rovers and probes collect data from planets such as Mars and how these findings contribute to understanding Earth's formation, environment, and processes; and 18.explain how scientists estimate the scale of the universe using ground-based and space-based

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		technologies (e.g., telescopes and measurements of redshift and standard candles).
Performance Standard <i>The learners can design and communicate simple, evidence-based explanations or proposals that demonstrate understanding of environmental sustainability, climate change, and the role of space technologies in monitoring Earth and improving daily life.</i>		
Suggested Performance Tasks • Design a simple sustainability plan or campaign for their home, school, or community (e.g., waste reduction, energy saving, or water conservation). • Create an infographic or report showing causes of climate change, effects on local communities, and possible solutions. • Develop a presentation illustrating how satellites collect and transmit data and explaining the role of space technologies in everyday applications such as weather forecasting, disaster preparedness, and communication.		